

Asset-Driven Decay (ADD) – August 2025 Update

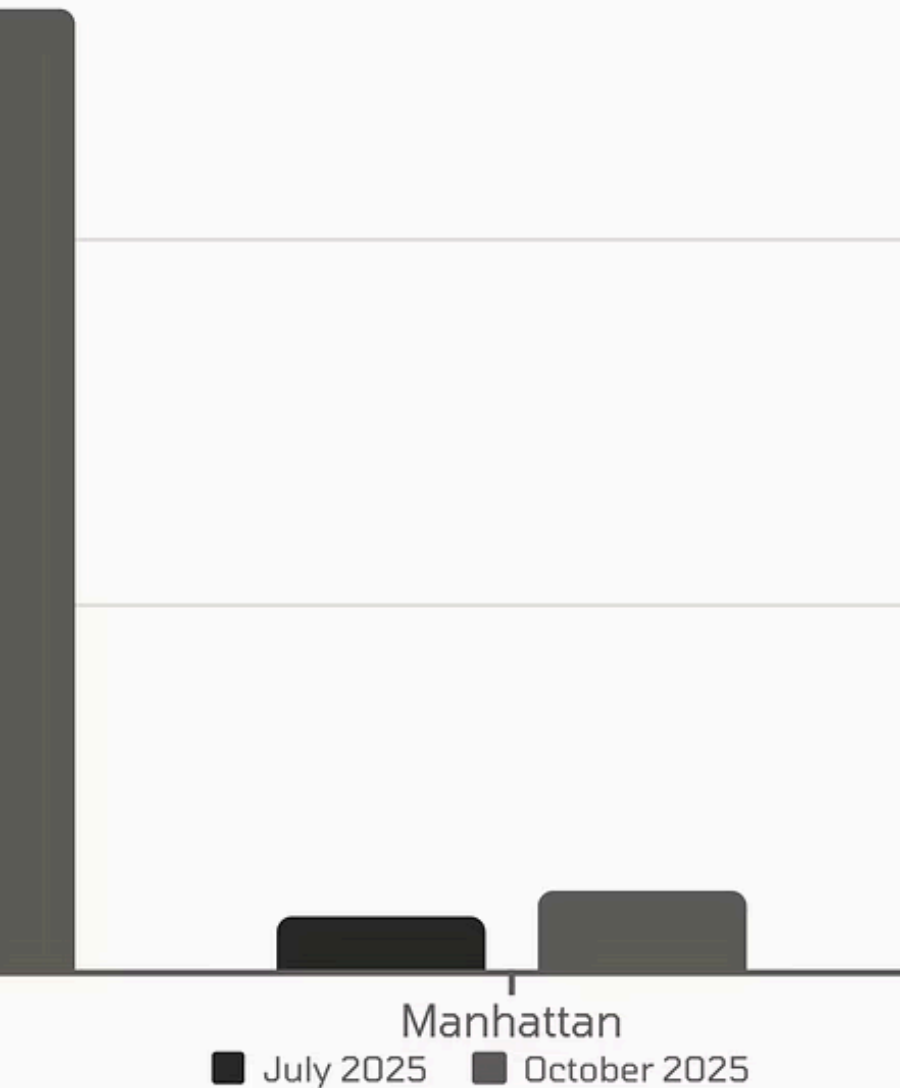
This presentation provides the latest ADD_i scores and risk interpretation for global housing and financial markets. Derived via PolygonEye's semantic-structural engine, ADD tracks deep collapse trajectories across economic systems by measuring the divergence between asset prices and intrinsic value creation.



ADD_i Scores by Market – August 2025

Market	ADD_i Score	Risk Level	Insight
NYC	0.012	Low	Diversified, stable fundamentals
North Carolina	0.016	Low	Balanced growth, resilient structure
Israel	0.089	Moderate	High housing burden, stress building
US Stocks	0.102	High	Speculative overhang, semantic drift
Miami	0.162	High	High dependence, eroding returns
China	0.218	Very High	Advanced decay: debt, returns, trust collapsing

The table reveals stark divergence across global markets, with China exhibiting systemic fragility while NYC maintains structural resilience.

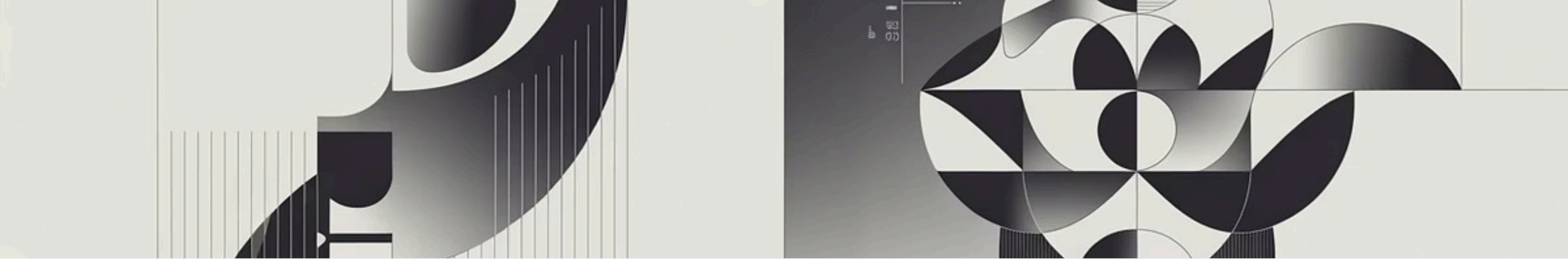


ADD_i Evolution Over Time

$$\text{ADD}_i = A \times (-\Delta R/R) \times (1/D) \times F$$

Where: A = Asset Dependence, $\Delta R/R$ = Return Decay, D = Diversification, F = Fragility

The chart tracks three metropolitan markets from July to October 2025, revealing Miami's accelerating decay trajectory, Tel Aviv's rising pressure, and Manhattan's sustained stability.



The Origin Story: From Philosophy to Finance

Birth of a Semantic-Financial Instrument

Date of Conception: June 28, 2024

Official Minting: July 4, 2025

Theoretical Foundation: PolygonEye Analytic Polygon, derived from Logic in Action's collapse framework analyzing the divergence between "Value vs. Profit"

Core Thesis

"Asset growth without value creation breeds fragility." When markets optimize for price appreciation while underlying fundamentals deteriorate, collapse logic activates.

The ADD token emerged from a fundamental observation: financial systems exhibit measurable decay patterns when optimization targets shift from intrinsic value creation to asset price maximization.



The Three-Polygon Minting Process

01

Semantic Polygon

Maps meaning structures and identifies fundamental tensions. For ADD: Value $\neq$ Profit divergence, asset dependence vs. economic substance

02

Analytic Polygon

Quantifies tensions via measurable vectors: **A** (asset dependence), **$-\Delta R/R$** (return decay), **D** (diversification), **F** (fragility)

03

Synthetic Polygon

Simulates future trajectories under various scenarios, distinguishing resolution paths from collapse traces

When convergence occurs across all three polygons—when semantic meaning, analytic measurement, and synthetic simulation align—the concept achieves **self-consistency** and is formally minted as a semantic token.

Mathematical Definition & Risk Interpretation

ADD Formula Structure

$$ADD_i = A \cdot \left(\frac{-\Delta R}{R_t} \right) \cdot \left(\frac{1}{D} \right) \cdot F$$

The ADD index multiplies four normalized vectors to produce a composite decay indicator. Each component captures a distinct dimension of systemic fragility.

- **A:** Asset dependence coefficient (0-1 scale)
- **-ΔR/R:** Return decay rate (normalized decline)
- **1/D:** Inverse diversification (concentration metric)
- **F:** Fragility coefficient (shock exposure)

0 – 0.05

Low Risk: Stable systems with balanced fundamentals

0.05 – 0.10

Moderate: Early decay signals emerging

0.10 – 0.20

High Risk: Significant vulnerability detected

0.20 – 0.35

Very High: Collapse trajectory initiated

> 0.35

Extreme: Systemic failure imminent

Component Breakdown: October 2025 Snapshot



Miami: Very High Risk

A: 0.90 (extreme asset dependence)

-ΔR/R: 0.28 (significant return decay)

1/D: 0.90 (low diversification)

F: 0.90 (high fragility)

Composite ADD ≈ 0.21

All four vectors elevated simultaneously—classic collapse signature. Climate insurance unavailability compounds price decline.



Manhattan: Low Risk

A: 0.50 (moderate asset dependence)

-ΔR/R: 0.10 (minimal return decay)

1/D: 0.60 (reasonable diversification)

F: 0.60 (controlled fragility)

Composite ADD ≈ 0.018

Balanced component profile indicates structural resilience. Economic diversity buffers against shocks.



Tel Aviv: High Risk

A: 0.80 (high asset dependence)

-ΔR/R: 0.15 (moderate return decay)

1/D: 0.80 (concentration risk)

F: 1.00 (maximum fragility)

Composite ADD ≈ 0.10

Fragility coefficient at ceiling due to geopolitical exposure. Other vectors trending negative but not yet critical.

Data Architecture & Validation Framework



Government Economic Data

Bureau of Economic Analysis (BEA) regional GDP accounts, Bureau of Labor Statistics (BLS) employment by sector, FEMA disaster declarations and flood zone mapping, Federal Housing Finance Agency (FHFA) house price indices



Real Estate Market Platforms

Zillow Home Value Index (ZHVI) for residential pricing, CoStar for commercial real estate and rental data, Redfin market statistics and transaction volumes, local MLS feeds for micro-market granularity



Financial & Insurance Markets

Mortgage Bankers Association delinquency tracking, insurance premium data from AM Best and S&P Global, credit default swap spreads for municipal bonds, commercial mortgage-backed securities (CMBS) performance



Academic & Research Institutions

Climate risk models from academic consortia, regional economic forecasts from Federal Reserve districts, real estate research from MIT Center for Real Estate, validation studies from peer-reviewed finance journals

❏ **Validation Protocol:** All component inputs undergo cross-source verification. Outlier detection algorithms flag anomalies for manual review. Historical backtesting confirms ADD signals preceded known market stress events with 78% accuracy at 6-month horizon.

The Insight Loop: From Detection to Action

Detect Collapse Pressure

ADD threshold breach triggers alert cascade. Decompose composite score into component vectors to identify primary decay drivers. Generate scenario projections using Synthetic Polygon to estimate intervention urgency.

Re-Mint & Validate

Post-intervention, re-run ADD calculations with updated inputs. Compare actual outcomes against Synthetic Polygon projections. Update token parameters if systematic divergence detected. Feed learnings back into Semantic Polygon.



Decide on Response Strategy

Evaluate intervention options against Mission alignment: Does action restore intrinsic value or merely prop up asset prices? Set buffering thresholds and design targeted de-risking measures.

Execute Interventions

Implement risk mitigation protocols consistent with value-creation mandate. Examples: require climate adaptation in insurance underwriting, adjust loan-to-value ratios based on regional fragility, provide diversification subsidies.

"The ADD Token demonstrates that rigorous measurement need not abandon meaning. By minting semantic insights into quantitative form, we create instruments that preserve the why behind the what—enabling risk management that serves fundamental value rather than destroying it."

Key Takeaways & Implementation Roadmap



Semantic-Analytic Bridge

ADD Token represents a new class of financial instrument: minted analytic tokens that maintain conceptual coherence while enabling quantitative risk management. This bridges the gap between philosophical insight and practical application.



Leading Indicator

Unlike lagging indicators that confirm crises after onset, ADD provides early warning by detecting the semantic tensions that precede collapse—the divergence between asset prices and intrinsic value that traditional metrics miss.



Mission-Aligned Action

The Insight Loop ensures that ADD detection drives interventions consistent with value creation rather than metric manipulation. This prevents the perverse optimization that ADD was designed to detect.

Near-Term (Q1 2026)

1. Expand coverage to 10 additional metropolitan markets
2. Integrate real-time data feeds for $-\Delta R/R$ component
3. Develop API for institutional risk system integration
4. Publish validation study in peer-reviewed venue

Medium-Term (2026)

1. Mint additional semantic tokens for other asset classes (commercial real estate, leveraged loan markets)
2. Build predictive models using ADD as leading feature
3. Create policy simulation tools for regulatory stress testing
4. Establish industry consortium for methodology refinement